

Sex Hormone-binding Globulin

Test Details

METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Levels of SHBG are under the positive control of estrogens and thyroid hormones, and are suppressed by androgens. These influences dynamically control the liver synthesis of this carrier protein. Decreased levels of SHBG are frequently seen in hirsutism, virilization, obese postmenopausal women, and in women with diffuse hair loss. Increased levels may be present in cases of hyperthyroidism, testicular feminization, cirrhosis, male hypogonadism, pregnancy, women using oral contraceptives, and prepubertal children.

Special Instructions

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

Limitations

As with all tests containing monoclonal mouse antibodies, erroneous findings may be obtained from samples taken from patients who have been treated with monoclonal mouse antibodies or who have received them for diagnostic purposes.¹ In rare cases, interference due to extremely high titers of antibodies to streptavidin and ruthenium can occur.¹ The test contains additives that minimize these effects.

Custom Additional Information

Sex hormone-binding globulin (SHBG) is the blood transport protein for testosterone and estradiol. It is a large glycoprotein with a molecular weight of about 95 kD and exists as a homodimer composed of two identical subunits. Each subunit contains two disulfide bridges.²

Planar C¹⁸ and C¹⁹ steroids with a 17 α -hydroxyl group bind particularly well,^{3,4} whereas C¹⁹ 17-ketosteroids, such as dehydroepiandrosterone (DHEA) and androstenedione, do not bind so easily. SHBG has a high binding affinity to dihydrotestosterone (DHT), medium affinity to testosterone and estradiol, and only a low affinity to estrone, DHEA, androstenedione, and estriol.

SHBG binds reversibly to sexual steroids. Albumin, which exists in far higher concentrations than SHBG, also binds to sexual steroids—although with a clearly lower binding affinity (eg, about 100 times lower for testosterone).

SHBG has a half-life of about seven days and is produced mainly by the liver. Its synthesis and secretion are regulated by estrogen.^{5,6} SHBG serum concentrations depend on the extent, duration, and the kind of estrogen applied, and how regulation takes place. Androgens and gestagens with androgenic residual action have the opposite effect.

In serum, SHBG mainly takes over the transportation of steroids and the reduction/regulation of the effect of androgen.^{7,8} Decreased SHBG serum levels are associated with conditions in which elevated androgen levels are present or in which the effect of androgen on its target organs is excessive. This explains the gender-related differences seen between men and women, especially during puberty.

Measurement of SHBG can be an important indicator of an excessive/chronic androgenic action where androgen levels are normal, but where clinical symptoms would seem to indicate androgen in excess. SHBG is a useful supplementary parameter in the determination of androgen where a relatively high concentration of free androgen (eg, testosterone) is suspected.⁹

By calculating the free androgen index (FAI), also called free testosterone index (FTI), from the ratio of total testosterone (TT) to SHBG [$\% \text{ FAI or FTI} = (\text{TT} / \text{SHBG}) \times 100$], it is possible to calculate the approximate amount of free testosterone (FTc), as there is a direct correlation between FAI and FT. Only free testosterone is biologically active, and it best indicates the clinical situation of the patient. Free testosterone is also referred to as non-SHBG-bound testosterone and can be obtained by precipitation of the SHBG-bound-testosterone with ammonium sulfate, and by equilibrium dialysis.^{10,11}

Elevated SHBG levels can be seen in elderly men, and are often found in patients with hyperthyroidism and cirrhosis of the liver. SHBG levels also increase when oral contraceptives or antiepileptic drugs are taken.

Pregnant women have markedly higher SHBG serum concentrations due to their increased estrogen production. Decreased SHBG concentrations are often seen with hypothyroidism, polycystic ovarian syndrome (PCOS), obesity, hirsutism, elevated androgen levels, alopecia, and acromegaly.

Specimen Requirements

Specimen

Serum

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Gel-barrier tube (preferred) or red-top tube

Collection Instructions

If a red-top tube is used, transfer separated serum to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

- Male:

- 20-49 years: 16.5–55.9 nmol/L

- >49 years: 19.3–76.4 nmol/L

- Female:

- 20-49 years: 24.6–122.0 nmol/L

- >49 years: 17.3–125.0 nmol/L

Storage Instructions

Room temperature

Causes for Rejection

Citrate plasma specimen; EDTA plasma specimen; improper labeling

Footnotes

1. SHBG on Elecsys 1010/2010 and Modular Analytics E170, 2007-09, V 7 [package insert]. Indianapolis, Ind: Roche Diagnostics; 2007.

2. Petra PH. The plasma sex steroid binding protein (SBP or SHBG). A critical review of recent developments on the structure, molecular biology

and function. *J Steroid Biochem Molec Biol.* 1991; 40(4-6):735-753. PubMed 1958572 (<http://www.ncbi.nlm.nih.gov/pubmed/1958572>)

3. Avvakumov GV, Grishkovskaya I, Muller YA, Hammond GL. Resolution of the human sex hormone-binding globulin dimer interface and evidence for two steroid-binding sites per homodimer. *J Biol Chem.* 2001 Sep 14; 276(37):34453-34457. PubMed 11457864

(<http://www.ncbi.nlm.nih.gov/pubmed/11457864>)

4. Hammond GL, Bochinfuso WP. Sex hormone-binding globulin/androgen-binding protein: Steroid-binding and dimerization domains. *J Steroid Biochem Molec Biol.* 1995 Jun; 53(1-6):543-552. PubMed 7626508 (<http://www.ncbi.nlm.nih.gov/pubmed/7626508>)

5. Burger HG. Androgen production in women. *Fertil Steril.* 2002 Apr; 77(Suppl 4):S3-5. PubMed 12007895 (<http://www.ncbi.nlm.nih.gov/pubmed/12007895>)

6. Davis S. Testosterone deficiency in women. *J Reprod Med.* 2001 Mar; 46(3 Suppl):S291-296. PubMed 11304877

(<http://www.ncbi.nlm.nih.gov/pubmed/11304877>)

7. Rosner W, Hryb DJ, Khan MS, Nakhia AM, Romas NA. Sex hormone-binding globulin mediates steroid hormone signal transduction at the plasma membrane. *J Steroid Biochem Molec Biol.* 1999 Apr-Jun; 69(1-6):481-485. PubMed 10419028 (<http://www.ncbi.nlm.nih.gov/pubmed/10419028>)

8. Burger HG, Davis SR. The role of androgen therapy. *Best Pract Res Clin Endocrinol Metab.* 2003 Mar; 17(1):165-175. PubMed 12763519

(<http://www.ncbi.nlm.nih.gov/pubmed/12763519>)

9. Pugeat M, Crave JC, Tourniaire J, Forest MG. Clinical utility of sex hormone-binding globulin measurement. *Horm Res.* 1996; 45(3-5):148-155. PubMed 8964574 (<http://www.ncbi.nlm.nih.gov/pubmed/8964574>)

10. Vermeulen A, Verdonck L, Kaufman JM. A critical evaluation of simple methods for the estimation of free testosterone in serum. *J Clin Endocrinol Metab.* 1999 Oct; 84(10):3666-3672. PubMed 10523012

(<http://www.ncbi.nlm.nih.gov/pubmed/10523012>)

11. Morley JE, Patrick P, Perry HM 3rd. Evaluation of assays available to measure free testosterone. *Metabolism.* 2002 May; 51(5):554-559. PubMed 11979385 (<http://www.ncbi.nlm.nih.gov/pubmed/11979385>)

